

Krishnaa Vadivel | GPU Software Engineer | High-Performance Computing | Scientific ML

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Summary

GPU-focused computational scientist and software engineer building accelerator-enabled applications in Python and C++ for large-scale simulation and data-processing workloads. Experience spans CUDA, ROCm, MPI, OpenMP, national-lab supercomputing systems, and machine-learning models that accelerate scientific computing pipelines.

Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Developed distributed scientific software in Python and C++ for first-principles simulation workloads as part of an INCITE award using exascale resources, with MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc on Frontier at OLCF, Perlmutter at LBNL, and related national-lab clusters.
- Ported and optimized scientific-computing workloads for accelerator-enabled HPC environments, translating many-body and excited-state physics methods into scalable GPU-backed implementations.
- Built PyTorch and PyTorch Geometric models, including equivariant graph neural networks, to accelerate self-trapped-exciton calculations and related materials simulations.
- Packaged scientific software in Docker containers and used tools such as TotalView and Linaro DDT to debug distributed applications across heterogeneous compute environments.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS research in 2024, 2025, and 2026, grounding GPU and HPC work in production research workflows.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed CUDA-based software for high-volume acoustic waveform processing, visualization, and field-tool diagnostics in geological well logging systems.
- Built full-stack server applications with C#, ASP.NET, and Microsoft SQL Server for database design, real-time remote tool testing, and calibration of temperature and pressure sensors.
- Applied finite-element analysis with dolfinx for frequency analysis of steel pipes in well environments to support physics-driven engineering studies.

FindLight

Photonics Technical Writer

2018–2019

- Wrote clear technical content on lasers, optics, and photonics devices for engineering audiences and product-focused readers.

Selected Projects

Accelerator-Enabled Exciton-Phonon Simulation: Developed distributed first-principles calculations for excited-state and exciton-phonon problems with CUDA, ROCm, MPI, and OpenMP on leadership-class GPU clusters.

Scientific ML on GPU Systems: Developed PyTorch Geometric equivariant models to accelerate molecular and excitonic calculations in GPU-backed scientific workflows.

CUDA Containers for HPC Software: Packaged scientific applications in CUDA-ready Docker environments for portable deployment across local and HPC systems.

Matrix-Multiply AI Accelerator: Built a Verilog-based matrix-multiply accelerator project for AI-oriented datapath design and hardware-aware compute fundamentals.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Technical Skills

Programming: Python, C, C++, CUDA, JavaScript, Bash, SQL, C#

GPU / HPC: CUDA, ROCm, MPI, OpenMP, PETSc, SLEPc, TotalView, Linaro DDT, Frontier, Perlmutter

Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks

Systems: Linux, macOS, Windows, Docker, Docker Compose, Kubernetes, gcloud, CMake

Domains: GPU computing, high-performance computing, first-principles materials modeling, many-body theory, scientific software